



Data Converters

Chapter 06 – Comparators

Summer Term 2026
Dr.-Ing. Matthias Keller

UNI
FREIBURG

Laboratory for Microelectronics
Department of Microsystems Engineering – IMTEK
University of Freiburg



Acknowledgement

Concepts & slides are partially based on the lecture

VLSI Data Conversion Circuits

held by

Prof. B. Murmann, PhD, University of Hawai'i

during his sabbatical at the University of Freiburg in 2009



UNI
FREIBURG



Overview

- **Introduction**
- Design Considerations

26-06-18

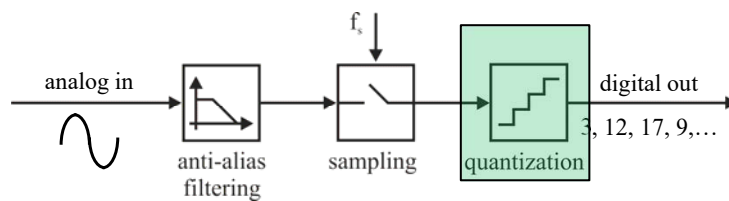
Data Converters

3

UNI
FREIBURG

Introduction

A/D conversion



Quantizers require circuit elements that “make decisions”.

most widely used “decision circuit”:
comparator

26-06-18

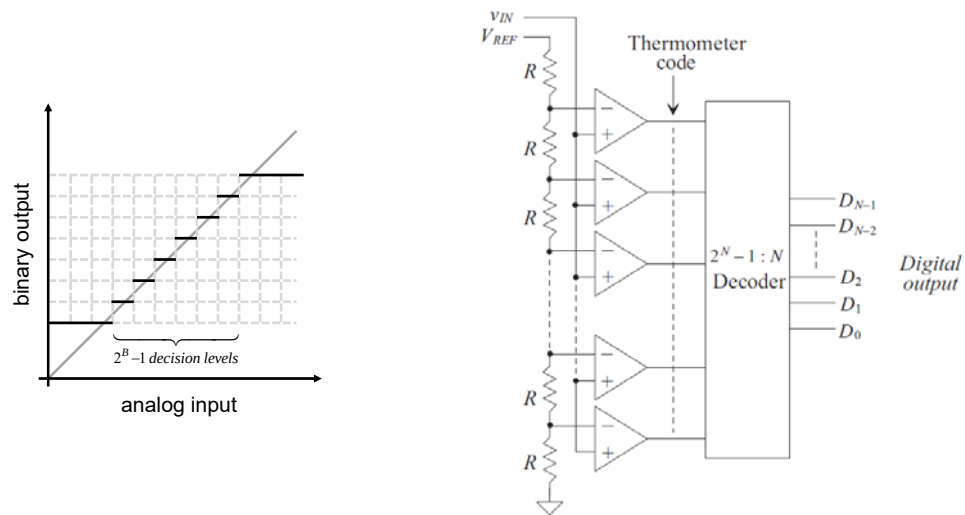
Data Converters

4

UNI
FREIBURG



Preview – Flash ADC



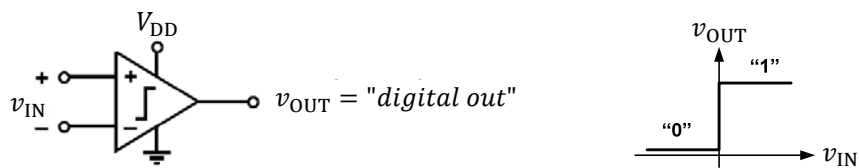
26-06-18

Data Converters

5

UNI
FREIBURG

Ideal Voltage Comparator



function:

Compare the instantaneous values of two analog voltages, e.g., an input signal and a reference voltage, and generate a digital 1 or 0 indicating the polarity of that difference.

26-06-18

Data Converters

6

UNI
FREIBURG



Design Considerations

- accuracy
 - gain (resolution)
 - offset
- speed
 - settling time (small-signal bandwidth)
 - delay time (slew rate)
 - overdrive recovery
- power dissipation
- input properties
 - sampled data versus continuous time
 - common-mode rejection
 - input capacitance and linearity of input capacitance
 - kickback noise

26-06-18

Data Converters

7

UNI
FREIBURG

Overview

- Introduction
- **Design Considerations**
 - **Accuracy & Speed**

26-06-18

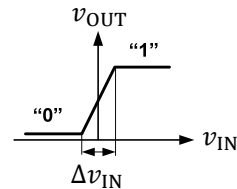
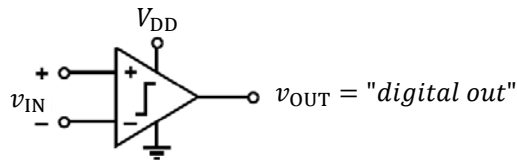
Data Converters

8

UNI
FREIBURG



Gain Requirements



$$\Delta v_{IN} = \frac{V_{DD}}{A_{V0}}$$

example: 12-bit ADC, $V_{DD} = 1.8 \text{ V}$, $FSR = 0.9 \text{ V} \rightarrow LSB = 0.9 \text{ V} / 4096$

Switching the comparator with half an $LSB \rightarrow A_{V0} = \frac{1.8 \text{ V}}{\frac{1}{2} \frac{0.9 \text{ V}}{4096}} \cong 16,000 \equiv 84 \text{ dB}$.

26-06-18

Data Converters

9

UNI
FREIBURG

How to Implement High Gain?

- Considerations
 - Amplification needs not be linear
 - Amplification needs not be continuous in time
(if comparator is used in a sampled data system)
→ clock signal will tell comparator when to make a decision
- Implementation options to be looked at
 - **single-stage amplification**
e.g. OTA or opamp in open-loop configuration
 - multi-stage amplification
e.g. cascade of resistively loaded differential pairs
 - regenerative latch using positive feedback
e.g. cross coupled inverters

26-06-18

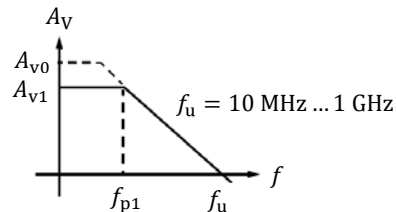
Data Converters

10

UNI
FREIBURG



Option I – Single-Stage Amplifier



$$f_{p1} = \frac{f_u}{A_{v1}} \approx \frac{1 \text{ GHz}}{16,000} = 62.5 \text{ kHz} \quad \tau_1 = \frac{1}{2\pi f_{p1}} = 2.5 \mu\text{s}$$

Too slow!

26-06-18

Data Converters

11

UNI
FREIBURG

How to Implement High Gain?

- Considerations
 - Amplification need not be linear
 - Amplification need not be continuous in time (if comparator is used in a sampled data system) → clock signal will tell comparator when to make a decision
- Implementation options to be looked at
 - single-stage amplification
 - e.g. OTA or opamp in open-loop configuration
 - **multi-stage amplification**
 - e.g. cascade of resistively loaded differential pairs**
 - regenerative latch using positive feedback
 - e.g. cross coupled inverters

26-06-18

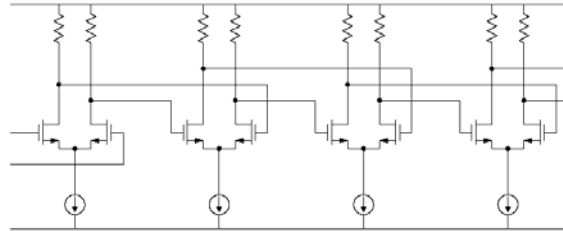
Data Converters

12

UNI
FREIBURG



Option II – Cascade of Open-Loop Amplifiers



per stage: $A_v = g_m R$ $\omega_p = \frac{1}{RC} = \frac{1}{RC_{GS}}$ $\omega_u = A_v \omega_p = \frac{g_m}{C_{GS}}$

possible options for a given, constant overall gain objective

- few stages with moderate gain
- lots of stages with low gain

26-06-18

Data Converters

13

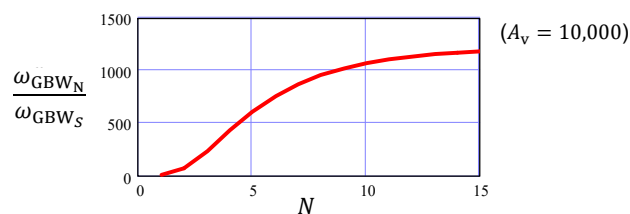
UNI
FREIBURG

Option II – Bandwidth of Cascade

intuition: cascade many low-gain stages $\rightarrow$ reduce $R \rightarrow$ increases bandwidth

analysis: $\frac{\omega_{GBW_N}}{\omega_{GBW_S}} = A_v^{(N-1)/N} \sqrt{2^{1/N} - 1}$

A_v – gain requirement
 N – number of stages
 ω_{GBW_S} – -3dB bandwidth of single stage
 ω_{GBW_N} – -3dB bandwidth of cascade



However, how do different orders N of the cascade perform relative to each other for a given overall gain when a step is applied?

Overall, increasing the order N increases the time it takes the signal to reach the output = delay time.

26-06-18

Data Converters

14

UNI
FREIBURG



Cascade of Amplifiers – Step Response

Let's consider the step response of a cascade consisting of N identical open-loop amplifiers, each modelled by the same single-pole transfer function.

The gain of the cascade is A_v , i.e., the gain per stage results in $\sqrt[N]{A_v}$.

$$\text{Moreover } p_{\text{stage}} A_{v,\text{stage}} = \omega_{u,\text{stage}} \rightarrow p_{\text{stage}} = \frac{\omega_{u,\text{stage}}}{\sqrt[N]{A_v}} = \frac{1}{\tau_{u,\text{stage}} \sqrt[N]{A_v}}$$

$$\frac{v_{\text{OUT}}(s)}{v_{\text{IN}}(s)} = \left(\frac{\sqrt[N]{A_v}}{1 + s \tau_{u,\text{stage}} \sqrt[N]{A_v}} \right)^N = \frac{A_v}{(1 + s \tau_{u,\text{stage}} \sqrt[N]{A_v})^N} \rightarrow v_{\text{OUT}}(s) = \frac{A_v}{(1 + s \tau_{u,\text{stage}} \sqrt[N]{A_v})^N} \frac{V_{\text{step}}}{s}$$

$$v_{\text{OUT}}(t) = \left(1 - e^{-\frac{t}{\tau_{u,\text{stage}} \sqrt[N]{A_v}}} \sum_{i=0}^{N-1} \frac{\left(\frac{t}{\tau_{u,\text{stage}} \sqrt[N]{A_v}} \right)^i}{i!} \right) A_v V_{\text{step}}$$

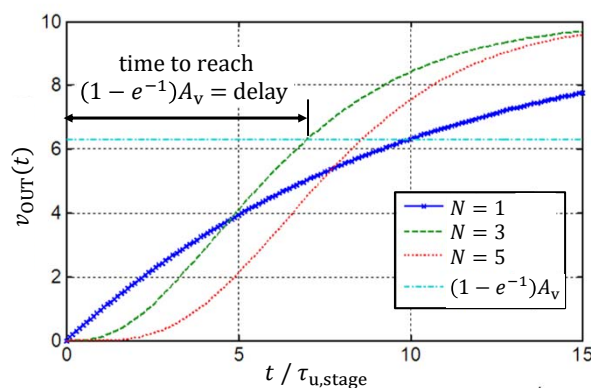
26-06-18

Data Converters

15

UNI
FREIBURG

Cascade of Amplifiers – Plot Step Response



For $A_v = 10$,
three stage amplifier wins!

Generally:
There seems to be an optimum number of stages w. r. t. the gain A_v .

$$v_{\text{OUT}}(t) = \left(1 - e^{-\frac{t}{\tau_{u,\text{stage}} \sqrt[N]{A_v}}} \sum_{i=0}^{N-1} \frac{\left(\frac{t}{\tau_{u,\text{stage}} \sqrt[N]{A_v}} \right)^i}{i!} \right) A_v V_{\text{step}}$$

26-06-18

Data Converters

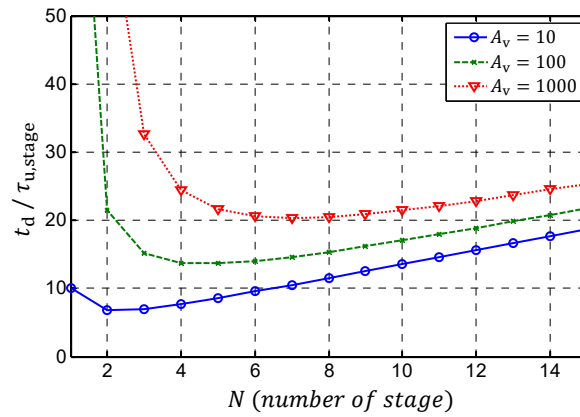
16

UNI
FREIBURG



Delay vs. Number of Stages

$V_{OUT}(t_d) = A_v(1 - e^{-1})$ solved numerically for t_d .



Shallow minima!

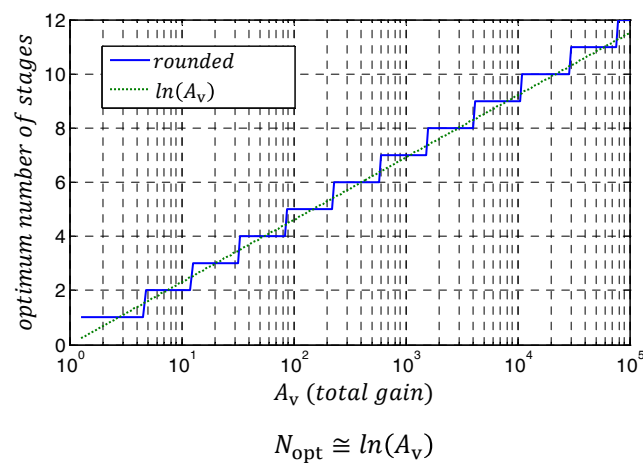
26-06-18

Data Converters

17

UNI
FREIBURG

Optimum Number of Stages



26-06-18

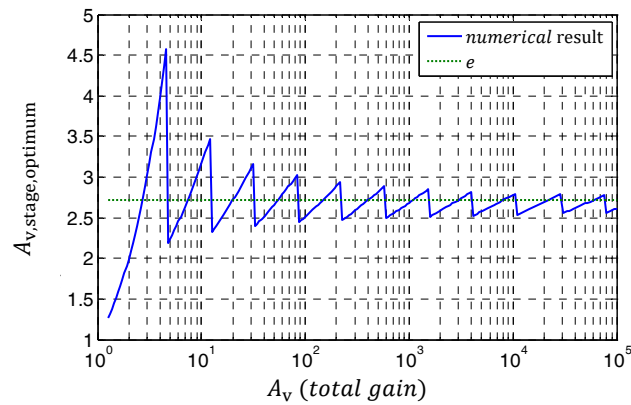
Data Converters

18

UNI
FREIBURG



Optimum Gain per Stage



$$N_{\text{opt}} \cong \ln(A_v) = \ln(A_{v,\text{stage,optimum}}^{N_{\text{opt}}}) \rightarrow e^{N_{\text{opt}}} \cong A_{v,\text{stage,optimum}}^{N_{\text{opt}}} \rightarrow A_{v,\text{stage,optimum}} \cong e$$

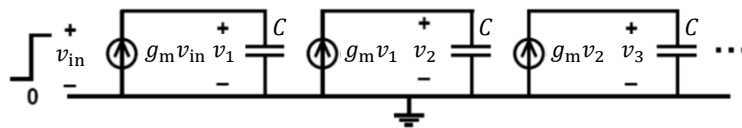
26-06-18

Data Converters

19

UNI
FREIBURG

Cascade of Integrators



intuition:

- load resistors in cascade of open loop amplifiers shunt current away from load capacitor $\rightarrow$ slows down amplification
- drop assumption $A_v = A_{v,\text{stage}}^N$ to see what happens...

analysis:

$$v_1 = \frac{g_m}{sC} v_{\text{in}} = \frac{\omega_u}{s} v_{\text{in}} \rightarrow v_N = v_{\text{out}} = \left(\frac{\omega_u}{s}\right)^N v_{\text{in}}$$

26-06-18

Data Converters

20

UNI
FREIBURG



Step Response – Comparison

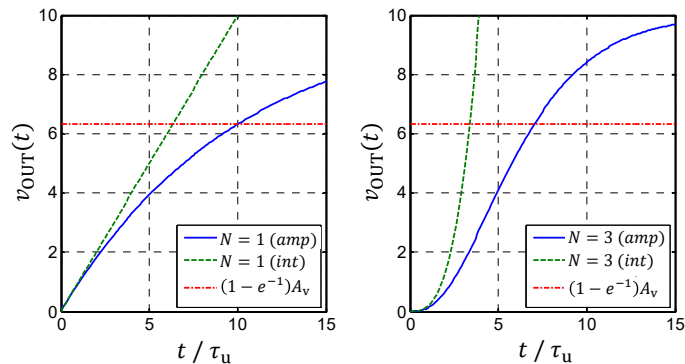
- higher amplification than cascade of resistively loaded amplifiers
- delay time τ_d

$$\frac{t_d}{\tau_u} = \left[N! \frac{v_{OUT}(\tau_d)}{V_{step}} \right]^{1/N}$$

- optimum number of stages N_{opt}

$$N_{opt} \approx 1.1 \ln \left[\frac{v_{OUT}(\tau_d)}{V_{step}} \right] + 0.79$$

- optimum gain per stage is close to $e = 2.7183 \dots$



$$v_{OUT}(s) = \left(\frac{\omega_u}{s} \right)^N \frac{V_{step}}{s} \quad \longleftrightarrow \quad v_{OUT}(t) = \frac{\omega_u^N}{N!} t^N V_{step}$$

26-06-18

Data Converters

21

UNI
FREIBURG

How to Implement High Gain?

- Considerations
 - Amplification need not be linear
 - Amplification need not be continuous in time (if comparator is used in a sampled data system) → clock signal will tell comparator when to make a decision
- Implementation options to be looked at
 - single-stage amplification
 - e.g. OTA or opamp in open-loop configuration
 - multi-stage amplification
 - e.g. cascade of resistively loaded differential pairs
 - **regenerative latch using positive feedback**
 - e.g. cross coupled inverters**

26-06-18

Data Converters

22

UNI
FREIBURG



Regenerative Sense Amplifier (Latch)

$t < 0$: set-up initial condition

$$v_{d0} = v_{10} - v_{20}$$

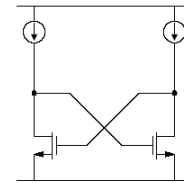
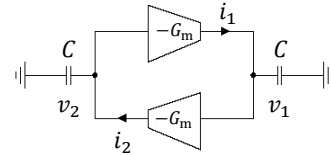
$t \geq 0$: enable positive feedback

$$\frac{dv_1}{dt} = \frac{i_1(t)}{C} = -\frac{g_m v_2(t)}{C} = -\frac{v_2(t)}{\tau_u}$$

$$\frac{dv_2}{dt} = \frac{i_2(t)}{C} = -\frac{g_m v_1(t)}{C} = -\frac{v_1(t)}{\tau_u}$$

$$v_1(t) - v_2(t) = v_d(t) = v_{d0} e^{\frac{t}{\tau_u}}$$

$$\frac{v_{OUT}(t)}{v_{d0}} = e^{\frac{t}{\tau_u}}$$



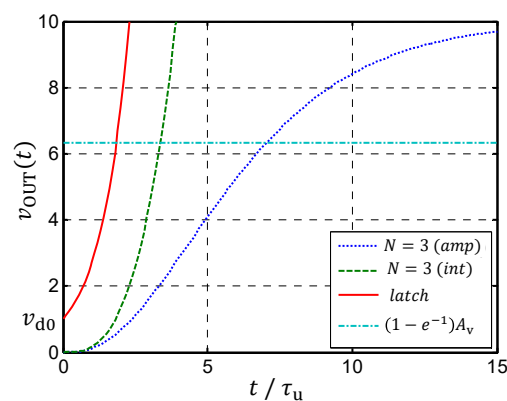
26-06-18

Data Converters

23

UNI
FREIBURG

Step Response – Comparison



Latch is much faster than cascade of amplifiers or integrators.

26-06-18

Data Converters

24

UNI
FREIBURG



„Gain“ of a Latch

$v_{OUT}(t_d)/v_{d0}$	t_d/τ_u
10	2.3
100	4.6
1,000	6.9
10,000	9.2

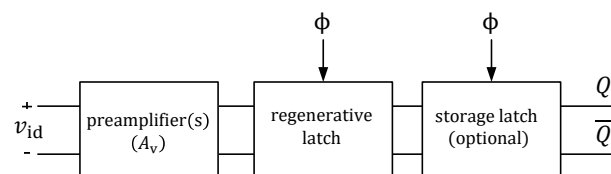
26-06-18

Data Converters

25

UNI
FREIBURG

Comparator – Typical Implementation



Why pre-amplification A_v ?

- reduction of metastability
- reduction of offset
 - hard to build latches with offset < 10 ... 100 mV
 - → use pre-amplification to lower input referred offset
- improvement of common-mode rejection
- attenuation of "kickback noise"

26-06-18

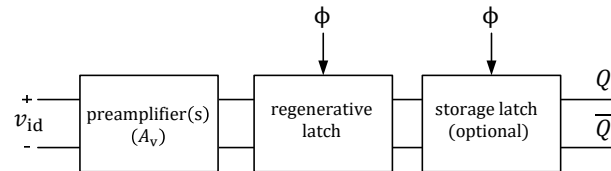
Data Converters

26

UNI
FREIBURG



Derivation Metastability I



Minimum initial latch input voltage needed to regenerate to V_{DD} within maximum available time t_{max} .

$$V_{DD} = v_{d0min} e^{\frac{t_{max}}{\tau_u}} \rightarrow v_{d0min} = \frac{V_{DD}}{e^{t_{max}/\tau_u}} \rightarrow \text{minimum pre-amplifier input } v_{id0min} = \frac{1}{A_v} \frac{V_{DD}}{e^{t_{max}/\tau_u}}$$

Any input signal $v_{id0} < v_{id0min}$ cannot be resolved $\rightarrow$ metastability!

Veendrick, IEEE JSSC 04/1980, Zojer, IEEE JSSC 06/1985

26-06-18

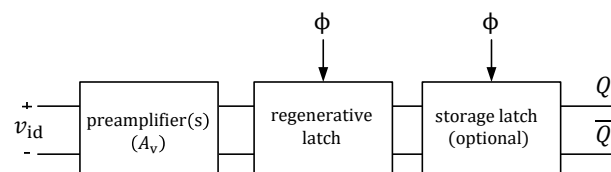
Data Converters

27

UNI
FREIBURG



Derivation Metastability II



assuming a uniform input signal distribution over the input signal range 0 V to v_{id0max}

$$P(error) = P(v_{id0} < v_{id0min}) = \frac{v_{id0min}}{v_{id0max}} = \frac{1}{A_v} \frac{V_{DD}}{v_{id0max}} e^{-t_{max}/\tau_u}$$

considering a B-bit Flash ADC

$$P(error) = \frac{1}{A_v} \frac{V_{DD}}{\frac{FSR}{2^B}} e^{-t_{max}/\tau_u}$$

26-06-18

Data Converters

28

UNI
FREIBURG



Example – 6-bit Flash ADC, $f_s = 500$ MHz

$$f_s = 500 \text{ MHz}, t_{\max} = T_s/2 = 1 \text{ ns}, \tau_u = 32 \text{ ps}, A_v = 3, FSR = 0.5 V_{DD}$$

$$P(\text{error}) = \frac{1}{A_v} \frac{V_{DD}}{FSR} e^{-t_{\max}/\tau_u} = \frac{2}{3} 2^6 e^{-1000/32} \cong 10^{-12}$$

mean time to failure MTF

$$MTF = \frac{1}{P(\text{error})f_s} = \frac{1}{10^{-12} \cdot 0.5 \cdot 10^9} = 2000 \text{ s} \cong 33 \text{ min}$$

ideally design for $MTF > 1 \dots 10$ years (if possible)

improve MTF by

- reducing speed/better technology $\rightarrow$ exponential dependence $e^{-t_{\max}/\tau_u}$
 - adding pre-amplification $\rightarrow$ linear dependence $1/A_v$

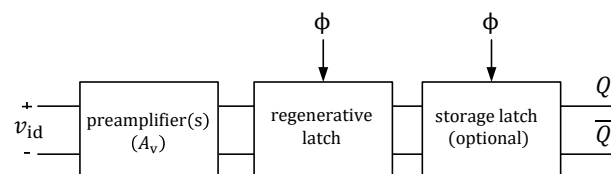
26-06-18

Data Converters

29

UNI
FREIBURG

Typical Implementation



Why pre-amplification A_v ?

- reduction of metastability
- **reduction of offset**
 - **hard to build latches with offset $< 10 \dots 100$ mV**
 - **$\rightarrow$ use pre-amplification to lower input referred offset**
- improvement of common-mode rejection
- attenuation of "kickback noise"

26-06-18

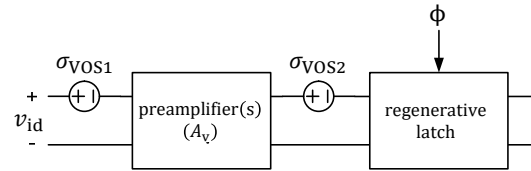
Data Converters

30

UNI
FREIBURG



Input Referred Offset



$$\sigma_{VOS}^2 = \sigma_{VOS1}^2 + \frac{1}{A_v^2} \sigma_{VOS2}^2$$

example: $\sigma_{VOS1} = 3 \text{ mV}$, $\sigma_{VOS2} = 30 \text{ mV}$, $A_v = 10$

$$\sigma_{VOS} = \sqrt{(3 \text{ mV})^2 + \frac{1}{10^2} (30 \text{ mV})^2} = 4.2 \text{ mV}$$

Offset of latch highly reduced by means of pre-amplification.

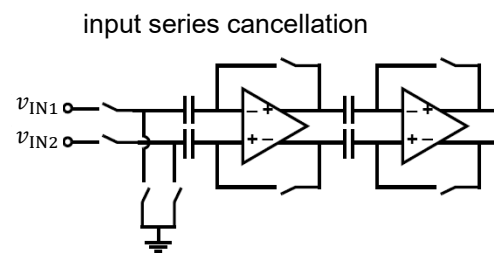
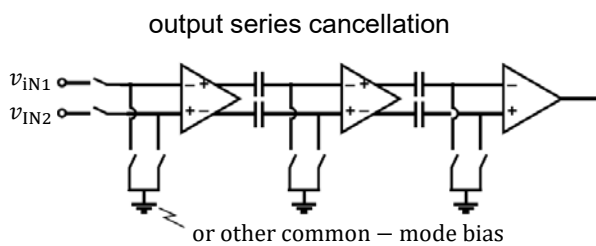
26-06-18

Data Converters

31

UNI
FREIBURG

Offset Compensation – Approaches



- Store offset voltage of each amplifier on a capacitor in a calibration phase.
 - Subtract it during regular operation.

26-06-18

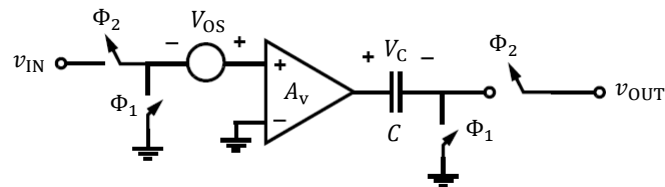
Data Converters

32

UNI
FREIBURG



Output Series Cancellation



phase 1: offset storage, phase 2: amplification

design considerations:

- ensure that amplifier does not saturate during phase 1
- make C sufficiently large to avoid attenuation and mitigate charge injection error

Poujois et al., IEEE JSSC, 08/1978

26-06-18

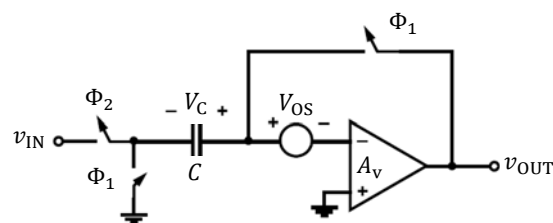
Data Converters

33

UNI
FREIBURG



Input Series Cancellation



phase 1: offset storage, phase 2: amplification

input referred offset is reduced to $V_{OS}/(A_v + 1)$ in phase 2
 $\rightarrow 4\times$ reduction for $A_v = 3$

26-06-18

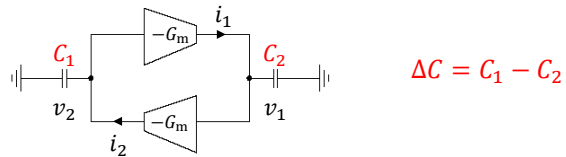
Data Converters

34

UNI
FREIBURG



Dynamic Offset



$C_1 \neq C_2$ causes dynamic, signal dependent offset

$$V_{OS} \cong 0.5 \frac{\Delta C}{C} \{V(t=0) - V(t)\}$$

example

$$V_{OS} \cong 0.5 \frac{10 \text{ fF}}{100 \text{ fF}} (1 \text{ V} - 0.5 \text{ V}) = 25 \text{ mV}$$

Nikoozadeh & Murmann, IEEE TCAS II, 12/2006

26-06-18

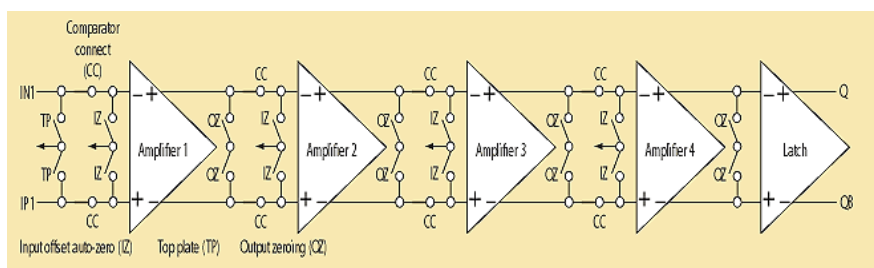
Data Converters

35

UNI
FREIBURG



Commercial Example – AD7671



16-bit, 1 MS/s successive approximation ADC, 0.6 μm CMOS technology

cascaded input series offset cancellation (4x pre-amplifiers)

offset < 3 mV (over process, temperature)

<http://www.elecdesign.com/Articles/Index.cfm?ArticleID=3956>

26-06-18

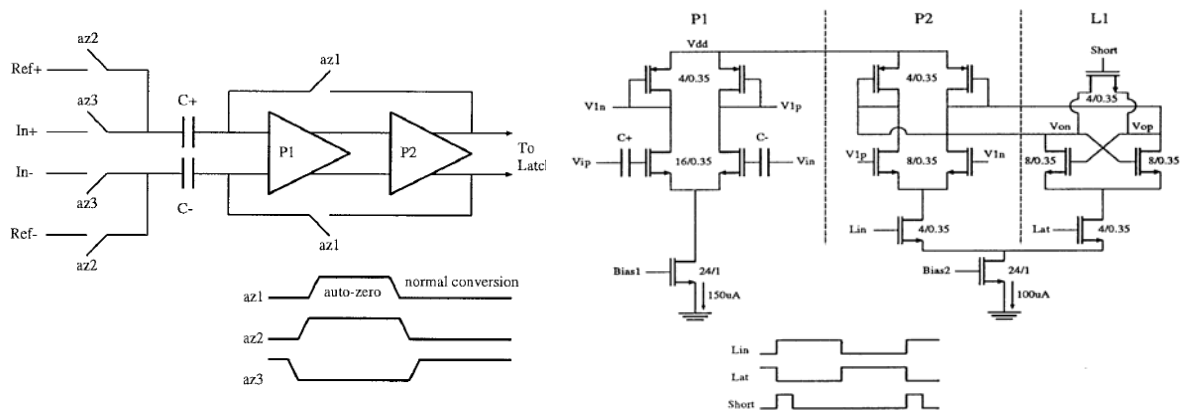
Data Converters

36

UNI
FREIBURG



Example I – Comparator with Input Series Cancellation



Mehr & Dalton, "A 500-MSample/s, 6-bit Nyquist-Rate ADC for Disk-Drive Read-Channel Applications,"
IEEE J. Solid-State Circuits, pp. 912-920, vol. 34, no. 7, July 1999

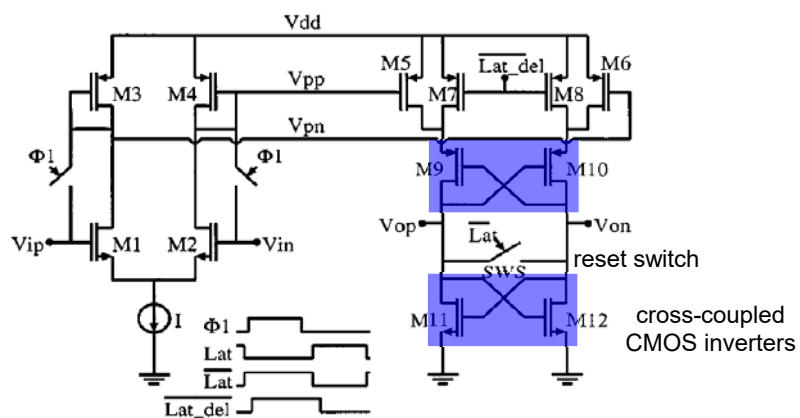
26-06-18

Data Converters

37

UNI
FREIBURG

Example II – Comparator with Input Series Cancellation



Mehr & Singer, "A 55-mW, 10-bit, 40-Msample/s Nyquist-Rate CMOS ADC,"
IEEE J. Solid-State Circuits, pp. 318-325, vol. 35, no. 3, March 2000

26-06-18

Data Converters

38

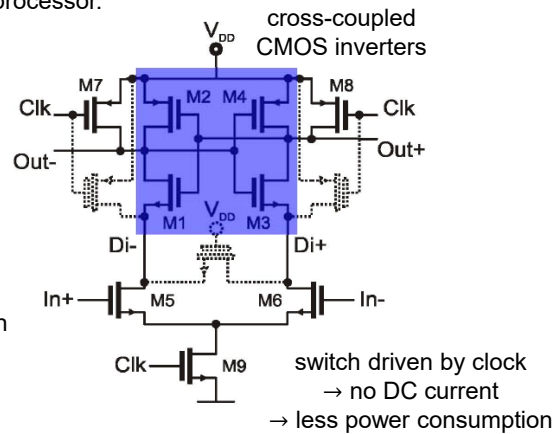
UNI
FREIBURG



Example III – Dynamic Comparator

Also called the StrongARM latch due to its first use as digital latch in the StrongARM microprocessor.

convert input voltages into currents & inject them into source of CMOS inverters in order to trigger the decision



26-06-18

Data Converters

39

UNI
FREIBURG

Example IV – Improved Dynamic Comparator

- less delay compared with StrongARM latch

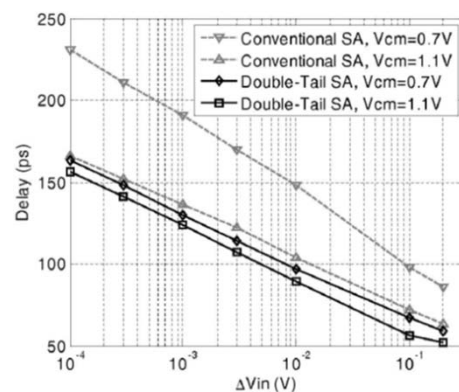
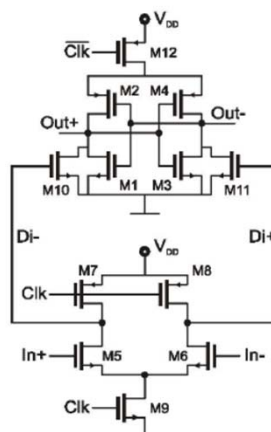


Figure 17.7.3: Simulated sense amplifier delays versus differential input voltage. The delay is the time between the clock edge and the instant when ΔOut crosses $1/2 V_{DD}$.

Schinkel et al., "A Double-Tail Latch-Type Voltage Sense Amplifier with 18ps Setup+Hold Time," IEEE Int. Solid-State Circuits Conf., 2007

26-06-18

Data Converters

40

UNI
FREIBURG